Motor Parameters

PWM Duty Cycle (%)

75

Supply Voltage (V)

12

PWM Frequency (Hz)

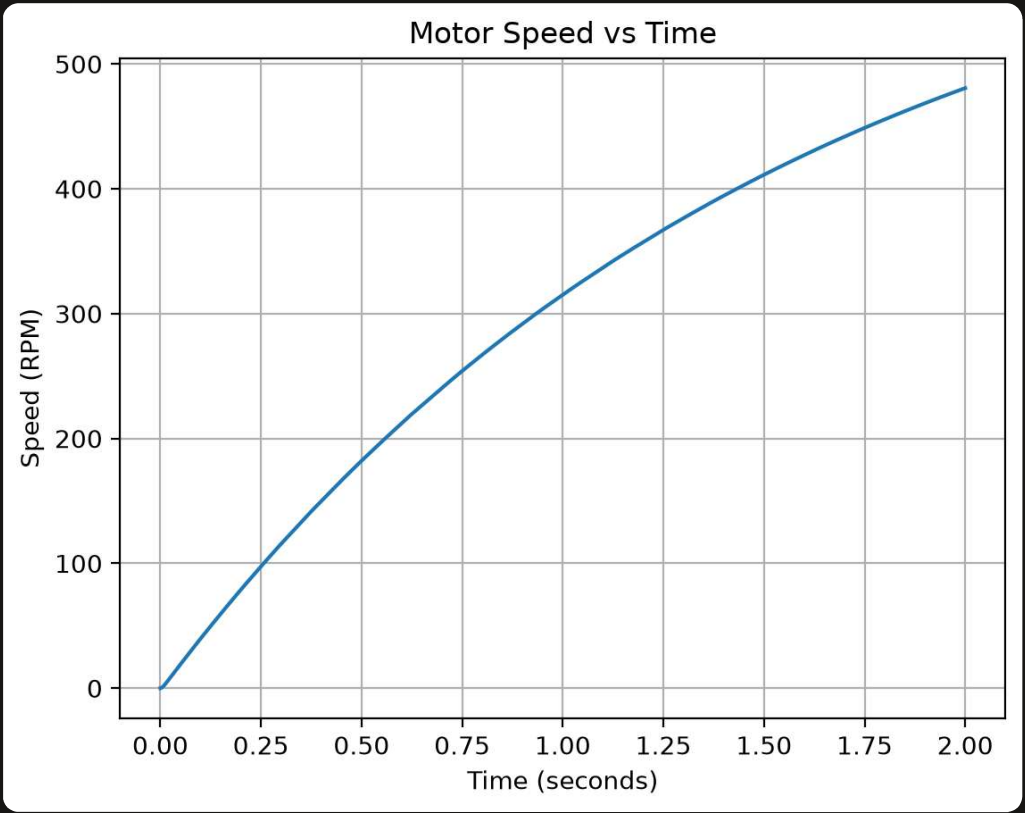
1000

DC Motor Speed Control Simulator

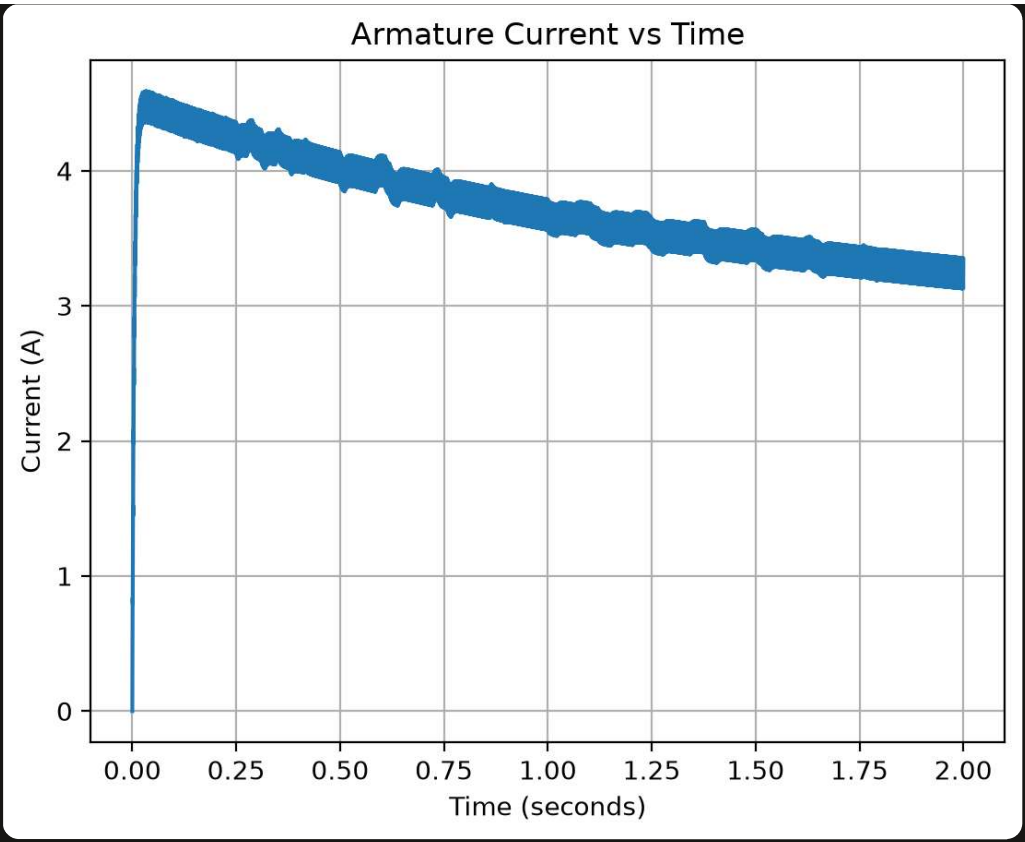
Simulating DC motor speed using Pulse Width Modulation (PWM).

Duty Cycle	Average Voltage	Motor Speed
75%	9.00 V	480.67 RPM

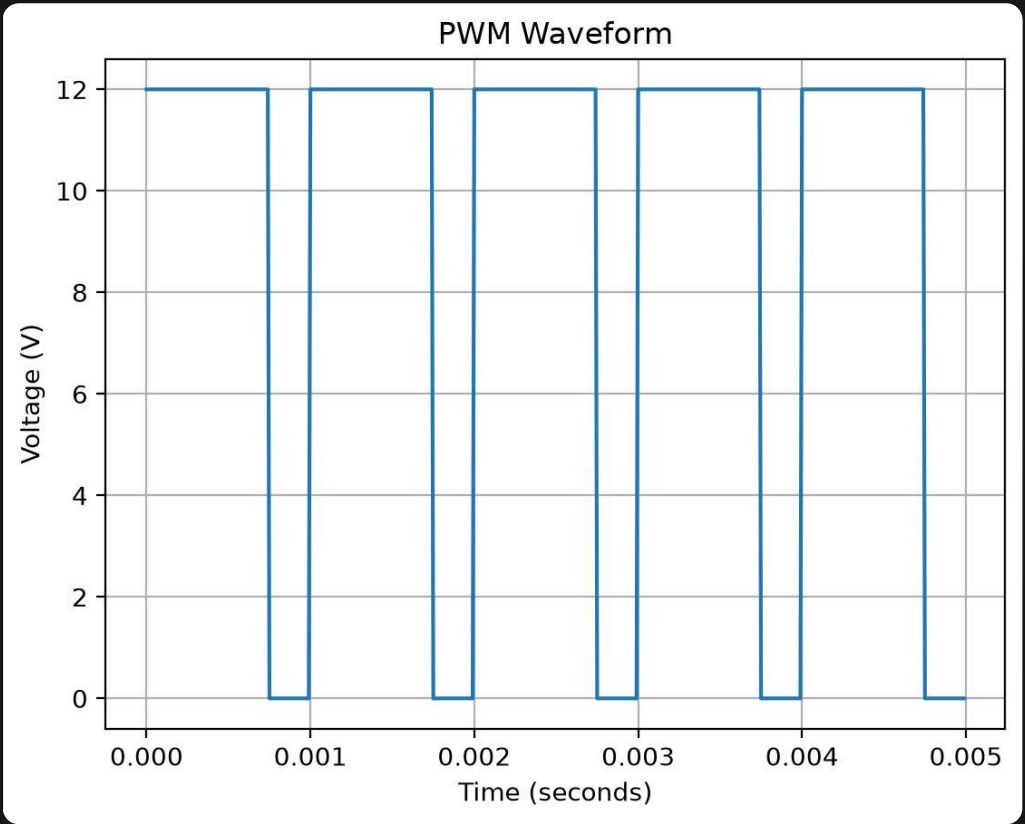
Motor Speed Response



Armature Current

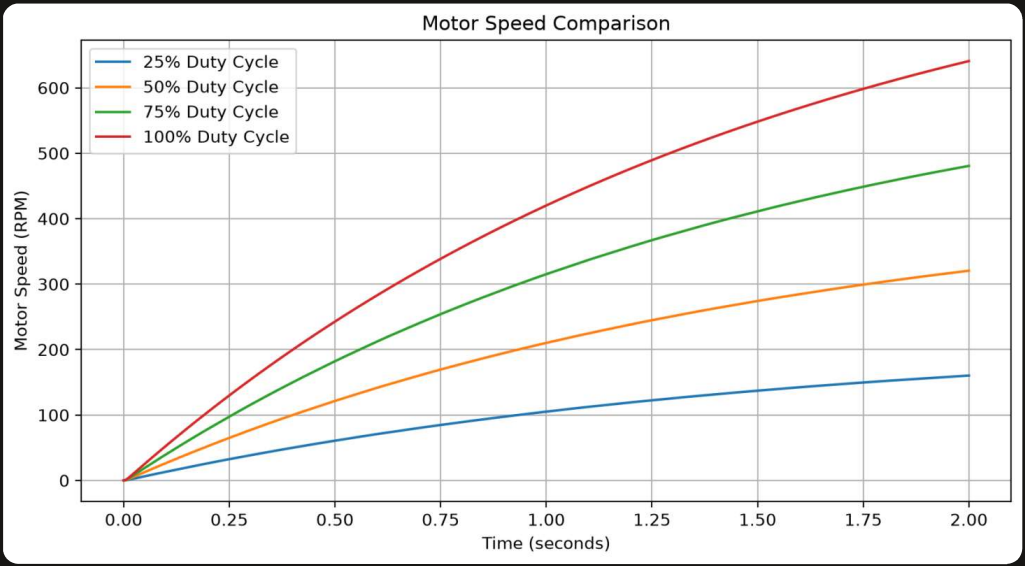


PWM Signal

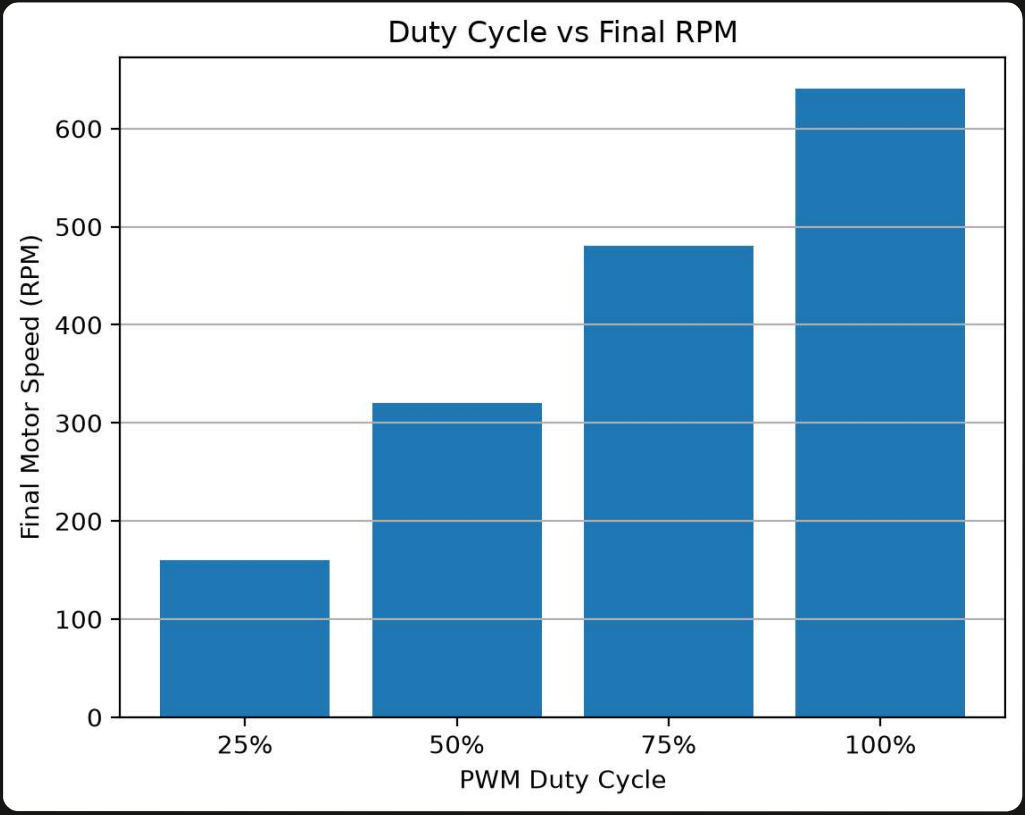


Simulation completed successfully!

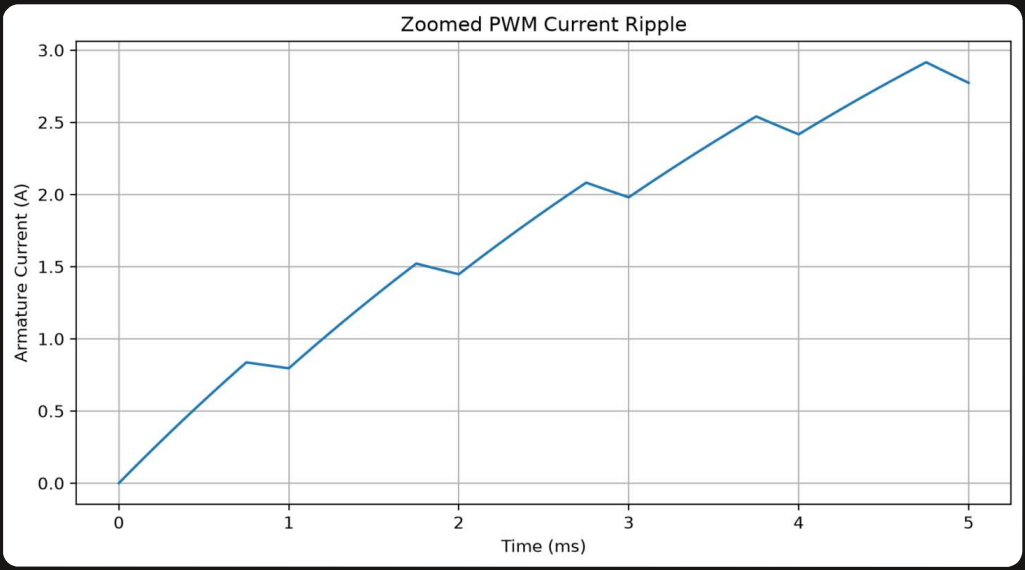
Duty Cycle vs Motor Speed Comparison



Final Speed at Different Duty Cycles



PWM Armature Current Ripple Analysis



Current Ripple (First 5 ms)

2.9157 A

Effect of Switching Frequency

